Math 133: Chain Rule and Quotient Rule

Instructor Key

1. Derivatives:

(a) $f'(x) = 15(3x-2)^4$.

(b) $u'(x) = \dfrac{2(x-3)-(2x+1)}{(x-3)^2} = \dfrac{-7}{(x-3)^2}$.

(c) $h'(x) = \dfrac{5}{2\sqrt{5x+4}}$.

(d) $v'(t) = \dfrac{2t(t+2)-(t^2+1)}{(t+2)^2} = \dfrac{t^2+4t-1}{(t+2)^2}$.

(e) $g'(t) = 3(t^2+4t+1)^2(2t+4)$.

(f) $p'(z) = -2(2z^2+3)^{-3}(4z) = \dfrac{-8z}{(2z^2+3)^3}$.

(g) $r'(x) = 4(x^3-2x+1)^3(3x^2-2)$.

(h) $m'(x) = \dfrac{x^2+1}{\sqrt{2x+1}} + 2x\sqrt{2x+1} = \dfrac{5x^2+2x+1}{\sqrt{2x+1}}$.

(i) $q'(z) = \dfrac{\frac{z+1}{2\sqrt{z+3}} - \sqrt{z+3}}{(z+1)^2} = \dfrac{-z-5}{2\sqrt{z+3}(z+1)^2}$.

(j) $s'(t) = \dfrac{3(t^2+1)^2(2t)}{2\sqrt{1+(t^2+1)^3}} = \dfrac{3t(t^2+1)^2}{\sqrt{1+(t^2+1)^3}}$.

2. $F(1) = (1^2+1)^2/(1+1) = 2$. Applying both rules gives

$$F'(x) = \frac{4x(x^2+1)(x+1)-(x^2+1)^2}{(x+1)^2}, \qquad F'(1) = \frac{16-4}{4} = 3.$$

The tangent line is $y-2 = 3(x-1)$, or $y = 3x-1$.